

Economic Espionage & Competitive Intelligence

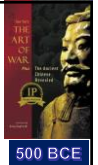


Department of Management Studies
Indian Institute of Technology Madras

LPS

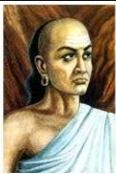
Sun Tzu: The Art of War

- When the 5 kinds of spies are all at work, none can discover the secret system.
- This is called "divine manipulation of the threads."
- It is the sovereign's most precious faculty.

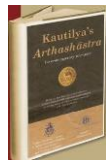


- local spies:** employing the services of the inhabitants of a district.
- inward spies:** making use of officials of the enemy.
- converted spies:** getting hold of the enemy's spies and using them for our own purposes.
- doomed spies:** doing certain things openly for purposes of deception, and allowing our spies to know of them and report them to the enemy.
- surviving spies:** those who bring back news from the enemy's camp.

War and Diplomacy in Ancient India



CHANAKYA aka KAUTILYA
350 - 275 BCE



- Kautilya **judged the means by the result**, and the result he sought was the general good of his kingdom.
- He was willing to use any possible means to **assassinate an enemy king**. Also arrange for the enemy to quarrel and fight among itself thereby defeating the enemy without full-scale battle.
- He wanted spies and secret agents to exploit the divisions within a country. Most countries have four kinds of **unhappy subjects - the enraged, the frightened, the greedy, and the proud**. Secret agents can widen and deepen these divisions by inciting these four types of people to act against their king.
- For him, all **ambassadors were potential spies** with diplomatic immunity.

Early Case of Industrial Espionage



L E T T R E
D U
PERE D'ENTRECOLLES,
Missionnaire de la Compagnie
de JESUS.

de son Ordre de la même Compagnie,
Prieur de l'abbaye de la Chine
à son Ordre.

A son Ordre, par le
Pere d'Entrecolles.

On Reverend Pere,
Le Pere de N. S. J.

Le Sieur que je fais de mon
en nom à King le salue pour les

1712. Lettres de quelques
l'abbaye de la Chine de son
Ordre, par le Pere d'Entrecolles,
Missionnaire de la Compagnie
de JESUS.

1712. Lettres de quelques
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de JESUS.



Porcelain of
Jingdezhen, China

The work of Father Francois Xavier d' Entrecolles (French Jesuit Priest)
- Revealed to Europe the manufacturing of Chinese porcelain in 1712.



"Teapot with Actresses" (porcelain painted in enamels)
Vezzi porcelain factory, Venice, Italy, 1725.

The Vezzi brothers were involved in a series of incidents of industrial espionage.



Production of porcelain at
Meissen (Germany)

Industrial Espionage: Britain and France (18th cent.)

- ❑ A large-scale **state-sponsored effort** to surreptitiously take British industrial technology to France.
- ❑ Witnesses confirmed both the inveigling of tradespersons abroad and the placing of **apprentices in England**.
- ❑ Protests by iron workers in Sheffield and steel workers in Newcastle about skilled industrial **workers being enticed abroad**.
- ❑ The **first English legislation** aimed at preventing economic and industrial espionage.

Snooping Gadgets



Camera



The Stunning Ring
(a beautiful self-defense
ring with pepper spray)



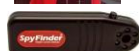
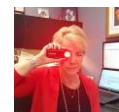
**Hidden Camera
Detector**



Mini Hidden camera
Wireless Monitoring System



**De-FIB-ulator
Voice Lie Detector**



British Airways Dirty Tricks Against Virgin Atlantic



- ❑ "Dirty Tricks" was the name of a business scandal targeted against Virgin Atlantic (VA).
- ❑ It was run by the rival airline British Airways (BA), whereby BA representatives would call up Virgin customers pretending to be from Virgin and try to get them to switch their flights to BA as well as hacking computers to gain access to insider information on Virgin's flights.
- ❑ The Virgin Group chairman, Richard Branson sued BA over it and was paid a total of **£3.51 million** in compensation, includes damages (\$945,000) and legal expenses (1993).

**British Air Tells Virgin Air It's Sorry
and Pays \$945,000**

The New York Times
Jan. 12, 1993

**When Richard Branson's airline won
\$945,000 from a lawsuit, he gave it
all to his employees**



Volkswagen Drives Away with General Motors' Secrets



Jose Ignacio Lopez, chief of production for GM's Opel division, left for a job with rival Volkswagen and he allegedly took a bundle of confidential documents with him. He also took seven of Opel's top executives.

GM claimed its trade secrets were used at Volkswagen.

The four-year legal battle was resolved in 1997.

The companies agreed on a settlement when GM said it would drop the lawsuits if the German automaker agreed to pay **\$100 million** and pledge to buy \$1 billion of GM car parts over seven years.

Oracle defends Microsoft spying



No Regrets

Oracle's chief Larry Ellison has defended the decision to spy on Microsoft. He admitted hiring private detectives to gain evidence of what it says is Microsoft's funding of public interest groups. Ellison said he had no regrets.



BBC News (29 June, 2000)

Oracle's aim was to discredit the campaign it says was bankrolled by Microsoft to persuade the government to drop its anti-trust case.

Microsoft is battling with the US government over plans to break-up the company because of violation of anti-trust laws.

The detective agency hired by Oracle, Investigative Group International, reportedly tried to obtain documents from the Association for Competitive Technology, a pro-Microsoft group, by paying staff who cleaned the group's Washington office.

Procter & Gamble Admits Spying on Rival Company, Unilever



P&G raided the garbage of rival Unilever for corporate secrets CINCINNATI, Ohio (Sep. 06, 2001)

Procter & Gamble (P&G) hired people to sift through the trash of rival firm Unilever (outside Chicago office) in order to obtain corporate secrets.

The espionage included paying people to pose as market analysts to glean information from P&G employees.

P&G will pay Unilever about **\$10 million** and agree to unusual third-party audit to settle the dispute.

Unilever had made demand to ensure that P&G did not change its marketing or product development plans for its hair care business after reviewing about 80 pages of confidential Unilever plans.

Ericsson Telecom: Spy Scandal

Ericsson dragged into spy scandal



The Ericsson affair could lead to a diplomatic row

BBC NEWS

Nov. 08, 2002

- ❑ Ericsson, the Swedish telecom company, being one of the leading suppliers of mobile phones, is also involved in radar and missile guidance systems for Sweden's Gripen fighter plane, the country's main strike aircraft.
- ❑ Three Ericsson employees were taken into custody suspected of passing on secret documents breaking company security rules.
- ❑ Sweden expelled two Russian diplomats who were said to be directly linked to the industrial espionage case at Ericsson.



U.S. Air Force Strips Boeing of \$1 B. in Potential Revenue



Boeing's use of stolen documents from rival Lockheed Martin

WASHINGTON D.C. (July 25, 2003)

The U.S. Air Force last week stripped Boeing Co. of [\\$1 billion](#) in planned launch contracts as part of a repudiation of the aerospace giant's use of stolen documents to outbid its main competitor.

The govt.'s action follows an investigation into Boeing's use of documents stolen from rival Lockheed Martin in a bid to win the bulk of a \$1.88 billion satellite launching contract in 1998.

Using as many as 25,000 stolen Lockheed documents, Boeing won the lion's share of the planned launches, taking home 19 launches to Lockheed's seven.

It took Boeing until last April to turn over all of the pilfered documents, which had been taken from Lockheed by a former worker.

HP – Spying on Dell?



A pretext for revenge

May 2007: H-P former VP Karl Kamb betrayed the company. He claims HP got his private phone records and spied on Dell. H-P denied all the charges..

Kamb inside the Las Vegas office of TV startup ByD:sign



Kamb persuaded then-CEO Carly Fiorina to take HP into the flat-panel-TV biz.



Three Tests For Industrial Espionage

- The **tactics** used to secure information are questionable since they appear to go beyond what might be deemed acceptable, ethical, or legal business practice.
- The **nature** of the information sought can itself be regarded as in some way private or confidential.
- The **purposes** for which the information is to be used are against the public interest.

Global Competitive Intelligence (CI) Activity: Some Facts

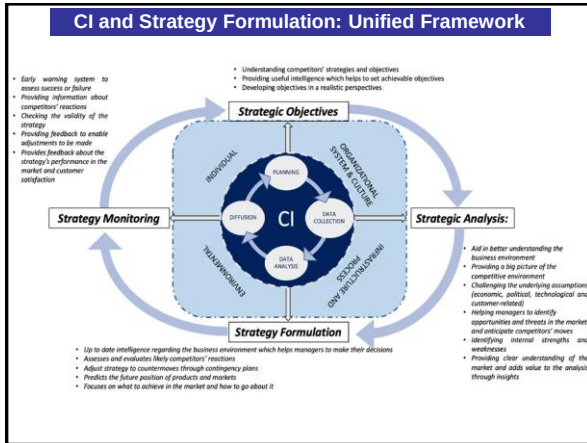
- ❑ 9 out of 10 multinational companies have a dedicated CI Unit
- ❑ 70% of recent Baldrige Award winners have had CI Units
- ❑ 25% of Fortune 500 Companies have dedicated CI Function at corporate / subsidiary level
- ❑ 80% of organized CI initiatives were conceived in the last decade



IBM's adoption of CI under Lou Gerstner played a key role in its historic turnaround



Microsoft's CI activity encompasses 100 FTEs aided by its CIKB web portal



What is CI ?	
Competitive Intelligence is...	Competitive Intelligence is not...
Information that has been analyzed to the point where you can make a decision.	Spying. Spying implies illegal or unethical activities. While spying does take place, it is a rare activity. Corporations do not want to find themselves in court, nor do they want to upset shareholders. For the most part, you will find spies in espionage novels, not in the executive suite.

What is CI ?	
Competitive Intelligence is...	Competitive Intelligence is not...
A tool to alert management to early warning of both threats and opportunities.	A crystal ball. There is no such thing as a true forecasting tool. Intelligence does give corporations good approximations of reality, near- and long-term. It does not predict the future.

What is CI ?	
Competitive Intelligence is...	Competitive Intelligence is not...
A means to deliver reasonable assessments. CI offers approximations and best views of the market and the competition. It is not a peek at the rival's financial books. Modern entrepreneurs want reasonable assessments. They don't expect every detail, just the best assessment at the time.	Database search. Databases offer just that — data. It is wonderful to have these remarkable tools. Nevertheless, databases do not massage or analyze the data. They certainly do not replace human beings who need to make decisions by examining the data and applying their common sense, experience, analytical tools, and intuition.

What is CI ?

Competitive Intelligence is...

Comes in many flavors. CI can mean many things to many people. A research scientist sees it as a heads-up on a competitor's new R&D initiatives.

A salesperson considers it insight on how his or her company should bid against another firm in order to win a contract.

A senior manager believes intelligence to be a long-term view on a marketplace and its rivals.

Competitive Intelligence is not...

The Internet or rumor chasing.

The Net is primarily a communications vehicle, not a deliverer of intelligence.

You can find hints at competitive strategy, but you will also uncover rumors disguised as fact, or speculation dressed up as reality.

Be wary of how you use or misuse the Net. Its reach is great, but you need to sift, sort, and be selective on its content.

What is CI ?

Competitive Intelligence is...

A way of life, a process. If a company uses CI correctly, it becomes a way of life for everyone in the corporation — not just the strategic planning or marketing staff.

It is a process by which critical information is available for anyone who needs it.

That process might be helped by computerization, but its success rests upon the people and their ability to use it.

Competitive Intelligence is not...

A job for one, smart person. A CEO might appoint one individual to oversee the CI process, but that one person cannot do it all.

At best, the CI Ringmaster, the coordinator of the program, keeps management informed and ensures that others in the organization become trained in ways to apply this tool within each of their SBUs.

What is CI ?

Competitive Intelligence is...

Both short- and long-term. A company can use intelligence for many immediate decisions, such as how to price a product or place an advertisement.

At the same time, you can use the same set of data to decide on long-term product development or market positioning.

Competitive Intelligence is not...

A spreadsheet. "If it's not a number, it's not intelligence." This is an unspoken, but often thought of, refrain among managers.

Intelligence comes in many forms, only one of which is a spreadsheet or some quantifiable result.

Management thinking, marketing strategy, and ability to innovate are only three among a host of issues that rely on a wide range of subjective, non-numeric intelligence.



CI: Definition

<http://www.scip.org/>

"Competitive intelligence (CI) is the process of monitoring the competitive environment.

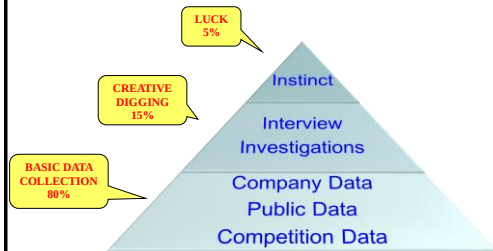
CI enables senior managers in companies of all sizes to make informed decisions about everything from marketing, R&D, and investing tactics to long-term business strategies.

Effective CI is a continuous process involving the legal and ethical collection of information, analysis that doesn't avoid unwelcome conclusions, and controlled dissemination of actionable intelligence to decision makers."

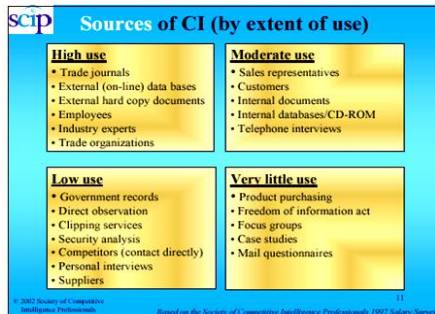
Code of CI Ethics – Dos and Don'ts

- ☑ Abide by all applicable laws and regulations of the land including company's Code of Conduct.
- ☑ If a competitor volunteers to share or inadvertently shares sensitive/confidential information, terminate info exchange politely.
- ☑ Adhere to all requests of confidentiality needs of clients/vendors.
- ☑ Maintain professional integrity by avoiding impersonation, misrepresentation and/or omission of pertinent data.
- ☑ Avoid conversation on pricing or contract-related info sharing with unrelated parties.
- ☑ Do not trespass the property (or Domain/ Network) of third party by subterfuge or covert means.
- ☑ Do not encourage anyone, including current and former competitor employees, to breach a contract or non-Disclosure obligation w.r.t. non-public information.
- ☑ Do not receive bid information like pricing policy, promotions, contracts, discounts or other sensitive information from a competitor directly.
- ☑ Do not affirmatively solicit Proprietary Information or Trade Secrets
- ☑ Do not hire / conduct job interviews in the pretext of knowing about the competitors / their business.
- ☑ Do not delegate the CI responsibilities to any associate not trained on the CI ethics.

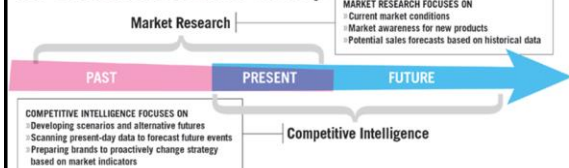
Competitive Intelligence Compilation



Competitive Intelligence Sources

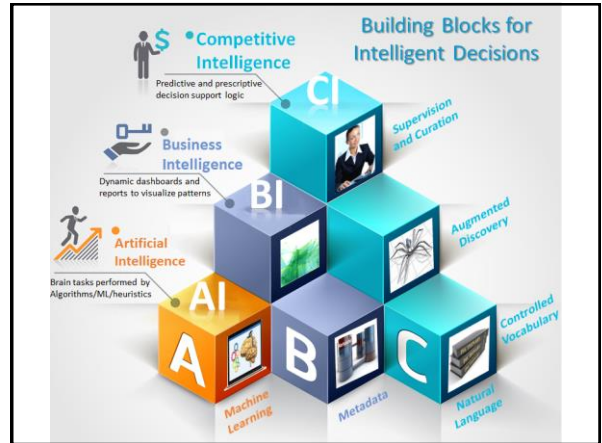


The CI-Market Research Overlap

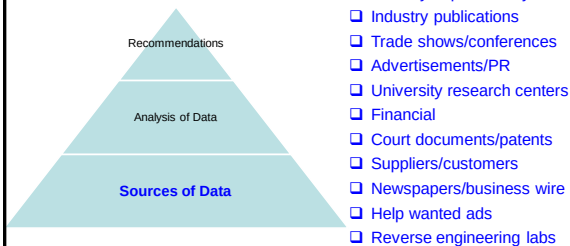


Business Intelligence vs Competitive Intelligence

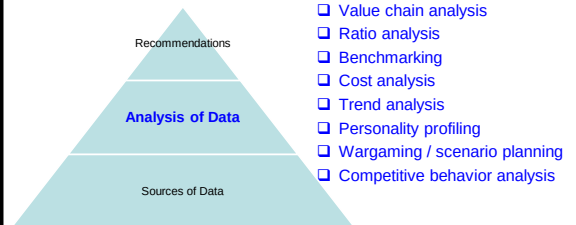
	BUSINESS INTELLIGENCE	COMPETITIVE INTELLIGENCE
Purpose	Analyzing mostly of the internal business processes, improvement operational and tactical decisions	Monitoring, analyzing of external environment - customers, suppliers, competitors, increasing strategic decisions
Scope	Organization (internal focus)	Environment of organization
Content / Data	Well structured content, internal data coming from data bases, ERP, transaction systems	Semistructured, unstructured content, external data coming from public, open resources, Internet
Used Tools, Technologies	OLAP, Data Mining, Data Warehouse	Advanced data mining, predictive modeling, web mining, text mining, agent technology, exponential random graph models, search based application, dashboards

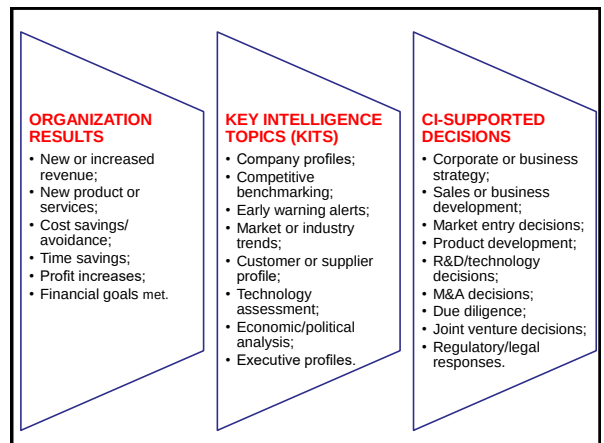
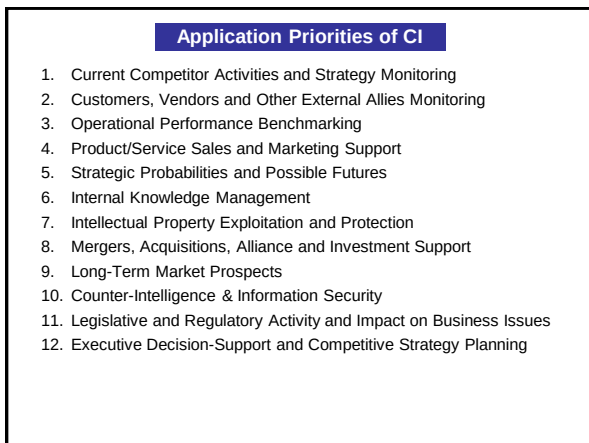
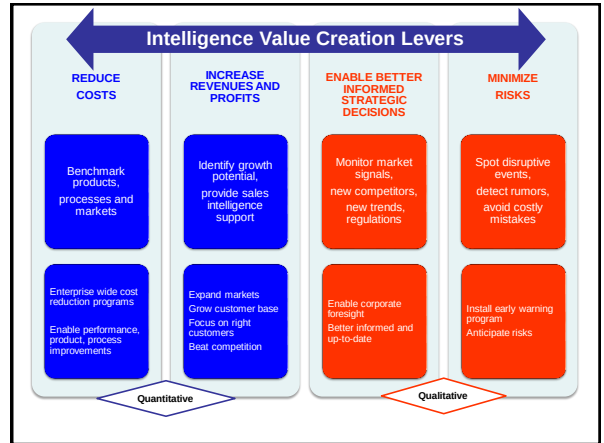
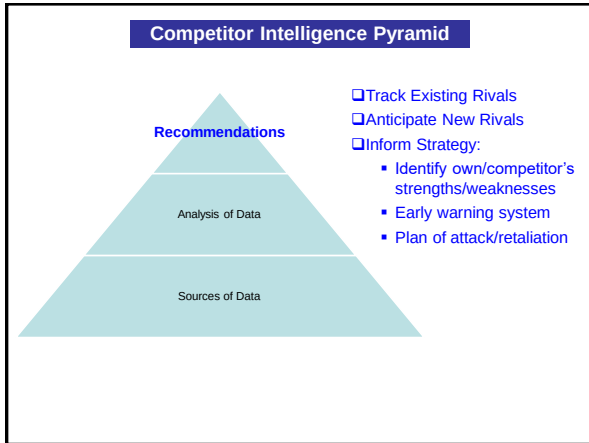


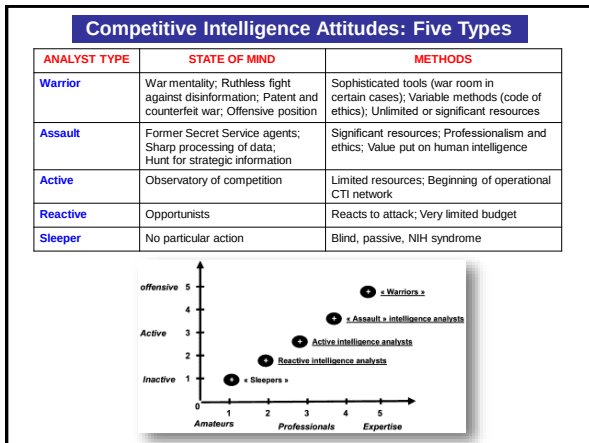
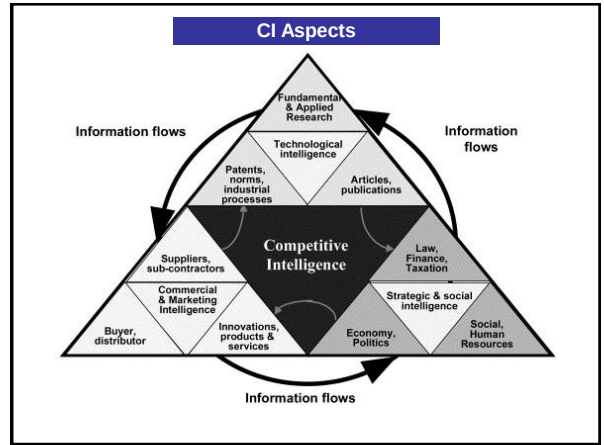
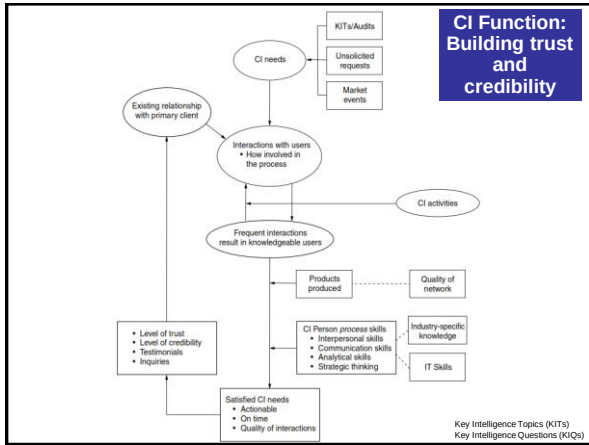
Competitor Intelligence Pyramid



Competitor Intelligence Pyramid



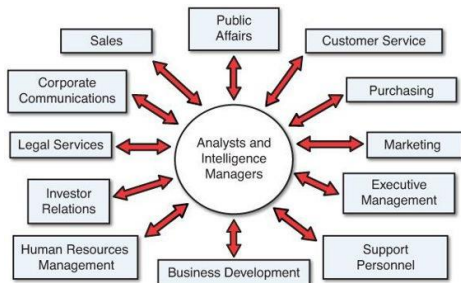




Intelligence Analysis at Differing Organizational Levels

Level	Brief Description	Typical Techniques Used
Strategic intelligence analysis (SIA)	Provides a framework for the other decision levels and are long-term, mostly infrequently taken, difficult to reverse, resource intense, and far reaching in nature. Also deals with competitive positioning issues.	- Opportunity, risk and threat assessments - Sector/competitor assessments - Trend analyses - Anomaly detection - Impact assessments
Tactical intelligence analysis (TIA)	Lacks macro-level analysis and micro-level focus on individual and specified matters. Supports decisions that are less pervasive than strategic decisions, typically involves formulating and implementing policies for the organization. Operates principally at the functional level.	- Cluster and pattern analyses - Stimulus-response analysis - Value constellation analysis
Operational intelligence analysis (OIA)	Support the day-to-day decisions needed to operate the organization. These decisions affect the organization for relatively shorter periods of time.	Many of these will be similar to operational products, usually focused on shorter time frames and more immediate activity.
Micro intelligence analysis (MIA)	Support the daily, customer-interfacing decisions needed to transact. These decisions individually impact the organization for relatively shorter periods of time, but cumulatively have a significant impact on profitability.	- Automation - Checklists - Decision trees - Metrification - Statistical analysis

Participants of Internal Networks



"F A R O U T" Approach

FAROUT is based on the premise that for analytical output to be insightful, intelligent, and valuable to business decision makers, it needs to meet a number of common characteristics. The output needs to be:

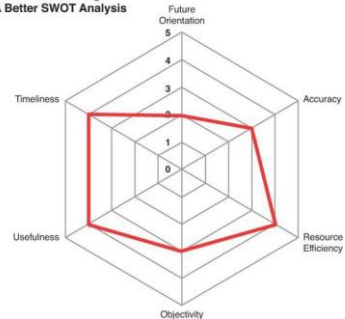
- ☐ Future-oriented
- ☐ Accurate
- ☐ Resource-efficient
- ☐ Objective
- ☐ Useful
- ☐ Timely

"F A R O U T" Approach: Scale

Factor	Scale
Future orientation	1 = output is not future-oriented. It may be too anchored in the past or present. 5 = the method is highly future-focused.
Accuracy	1 = the level of accuracy for outputs using this method is low, taking into account the probable sources of data underlying its application. 5 = the requirements of the model lead to the generation of highly accurate outputs.
Resource-efficiency	1 = this method requires a large volume of data, financial, and human resources, and is low in efficacy. 5 = this method is highly efficient in its use of resources and in deriving desired outputs from utilizing lesser inputs.
Objectivity	1 = provides low levels of objectivity due to its inability to reduce the presence of biases and mind-sets in its application. 5 = the potential for biases and distortions can be minimized through effective use of this method.
Usefulness	1 = application of the method delivers less useful (actionable) output and requires much additional work by or on behalf of a decision maker. 5 = provides a high level of valued and actionable output without additional effort by a decision maker.
Timeliness	1 = method that requires a great deal of time to properly complete. 5 = method takes little time to successfully complete.

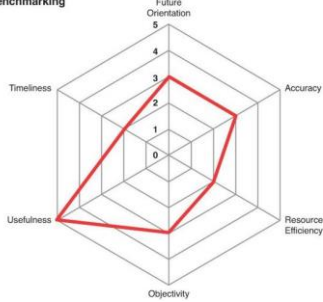
"F A R O U T" Approach: SWOT

FAROUT Screening:
A Better SWOT Analysis



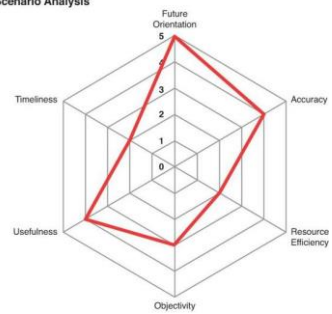
"F A R O U T" Approach: Benchmarking

FAROUT Screening:
Benchmarking



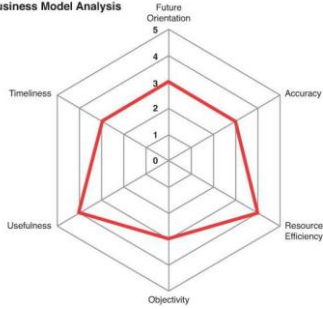
"F A R O U T" Approach: Scenario Analysis

FAROUT Screening:
Scenario Analysis



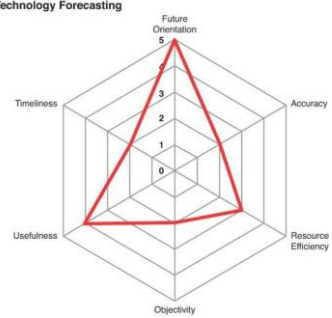
"F A R O U T" Approach: Business Model Analysis

FAROUT Screening:
Business Model Analysis



"F A R O U T" Approach: Technology Forecast

FAROUT Screening:
Technology Forecasting



Hierarchical Model of Analysis Failures

Level	Nature of Problem
Individual Analyst-Level Failures	■ Different natural analytical abilities ■ Naturally limited mental capacities ■ Natural motivation ■ Cognitive biases and perceptual distortion ■ Insufficient understanding and application of analysis tools and techniques
Analysis Task-Level Failures	■ Part of larger task ■ Task discontinuity ■ Unsatisfactory data inputs ■ Disconnects from decision making ■ Imbalance among key task facts
Internal Organizational-Level Failures	■ Some decision makers don't understand and appreciate analysis ■ Clients cannot specify their critical intelligence needs or questions ■ Under-resourcing the analysis function ■ Lack of analysis-specific IT support ■ Lack of thinking time ■ Organizational culture and politics ■ Time and trust ■ Invisibility and mystery ■ Misconception that everyone can do analysis
External Environment-Level Failures	■ Growing range of competitive factors ■ Complexity and turbulence ■ Data overload ■ Globalization ■ Educational deficiencies

Thank You

